MEASUREMENT AND INSTRUMENTATION

Q1. ……. are integrating instruments?

1. Ammeters
2. Voltmeters
3. Wattmeters
4. **Ampere-hour and Watt-hour meters**

Q2. Resistances can be measured with the help of a ……

1. Wattmeter
2. voltmeter
3. ammeter
4. **ohmmeter and resistance bridge**
5. all of the above

Q3. ……. instruments indicate the instantaneous value of the electrical quantity being measured at the time at which it is being measured?

1. Absolute
2. **Indicating**
3. Recording
4. Integrating

Q4. The use of ……. instruments is merely confined within laboratories as standardizing instruments.

1. **absolute**
2. indicating
3. recording
4. integrating
5. none of the above

Q5. ……. instruments measure the total quantity of electricity delivered at a particular time.

1. absolute
2. indicating
3. recording
4. **integrating**

Q6. According to application, instruments can be classified into ……. and …….

1. switch board
2. portable
3. **both 1 and 2**
4. moving coil
5. moving iron

6. both 4 and 5

Q7. The spring material used in a spring control device should have the following property:

1. should be non-magnetic
2. should have low-temperature co-efficient
3. should have low specific resistance
4. should not be subjected to fatigue
5. **all of the above**

Q8. Which of the following properties a damping oil must possess?

1. must be a good insulator
2. should be non-evaporating
3. should not have corrosive action upon the metal of the vane
4. the viscosity of the oil should not change with the temperature
5. **all of the above**

Q9. A ……. device prevents the oscillation of the moving system and enables the latter to reach its final position quickly.

1. deflecting
2. controlling
3. **damping**
4. all of the above

Q10. An induction meter can handle current upto …….

1. 10 A
2. 30 A
3. 60 A
4. **100 A**

Q11. For handling greater currents induction wattmeter are used in conjunction with …….

1. potential transformer
2. **current transformer**
3. power transformer
4. all of the above

Q12. …….. devices may be used for extending the range of the instruments?

1. shunts
2. multipliers
3. current transformers
4. potential transformer

5. **all of the above**

Q13. A moving-coil permanent-magnet instrument can be used as flux-meter by ……

1. using a low resistance shunt
2. using a high series resistance
3. **eliminating the control springs**
4. making control springs of large moment of inertia

Q14. A moving-coil permanent-magnet instrument can be used as …….. by using a low resistance shunt.

1. **ammeter**
2. voltmeter
3. flux-meter
4. ballistic galvanometer

Q15. A potentiometer may be used for ……

1. measurement of resistance
2. measurement of current
3. calibration of ammeter
4. calibration of voltmeter
5. **all of the above**

Q16. ……. are not used on D.C. circuits.

1. Mercury motor meters
2. commutator motor meters
3. **induction meters**
4. none of the above

Q17. ……. is an essential part of a motor meter .

1. an operating torque system
2. a barking device
3. revolution registering device
4. **all of the above**

Q18. Induction type single phase energy meters measure electric energy in ……..

1. kW
2. Wh
3. **kWh**
4. VAR
5. none of the above

Q19. The pointer of an indicating instrument should be ……

1. **very light**

2. very heavy
3. either 1 or 2
4. neither 1 nor 2

Q20. The household energy meter is …….

1. an indicating instrument
2. a recording instrument
3. **an integrating instrument**
4. none of the above

Q21. The chemical effect of current is used in …….

1. **D.C. ammeter hour meter**
2. D.D. ammeter
3. D.C. energy meter
4. none f the above

Q22. In majority of instruments, damping is provided by ……

1. fluid friction
2. spring
3. **eddy currents**
4. all of the above

Q23. An ammeter is a ……. instrument.

1. **secondary instrument**
2. absolute instrument
3. recording instrument
4. integrating instrument

Q24. The switchboard instruments should be mounted in ……. position.

1. **vertical**
2. horizontal
3. either 1 or 2
4. neither 1 nor 2

Q25. The function of shunt in an ammeter is to …….

1. **by pass the current**
2. increase the sensitivity of the ammeter
3. increase the resistance of ammeter
4. none of the above

Q26. The multiplier and the meter-coil in a voltmeter are in …….

1. **series**
2. parallel
3. series-parallel

4. none of the above

Q27. A moving iron instrument can be used for ……..

1. D.C. only
2. A.C. only
3. **both D.C. and A.C.**
4. none of the above

Q28. The scale of a rectifier instrument is ……..

1. **linear**
2. non-linear
3. either 1 or 2
4. neither 1 nor 2

Q29. To measure current at high frequency, we must use ……..

1. moving iron instrument
2. electrostatic instrument
3. **thermocouple instrument**
4. none of the above

Q30. The resistance in the circuit of the moving coil of a dynamometer should be ……

1. zero
2. low
3. **high**
4. none of the above

Q31. A dynamometer wattmeter can be used for ……

1. D.C. only
2. A.C. only
3. **both D.C. and A.C.**
4. none of the above

Q32. An induction wattmeter can be used for ……

1. **D.C. only**
2. A.C. only
3. both D.C. and A.C.
4. none of the above

Q33. In a 3-phase power measurement by two wattmeter method, the reading on one of the wattmeter is zero. The power factor of the load must be …….

1. unity
2. **0.5**
3. 0.3

4. zero

Q34. The adjustment of position of shading bands, in an energy meter is done to provide …….

1. **friction compensation**
2. creep compensation
3. braking torque
4. none of the above

Q35. An ohmmeter is a …… instrument.

1. moving iron
2. **moving coil**
3. dynamometer
4. none of the above

Q36. To measure a very high resistance, we should use ……..

1. Kelvin's double bridge
2. Wheat stone bridge
3. **Meggar**
4. none of the above

Q37. The electric power t a Meggar is provided by ……..

1. battery
2. **permanent magnet D.C. generator**
3. A.C. generator
4. any of the above

Q38. In a Meggar cntrolling torque is providec by …….

1. spring
2. gravity
3. **coil**
4. eddy current

Q39. The operating voltage of a Meggar is about …….V.

1. 6 V
2. 12 V
3. **40 V**
4. 100 V

Q40. Murray loop test can be used for location of …….

1. ground fault on a cable
2. short circuit fault on a cable
3. **both the ground fault and short-circuit fault**
4. none of the above

Q41. …… should be used for an accurate measurement of low D.C. voltage.

1. small range moving coil voltmeter
2. **D.C. potentiometer**
3. small range thermocouple voltmeter
4. none of the above

Q42. The best device to measure the true open circuit e.m.f. of a battery is …….

1. D.C. voltmetr
2. ammeter and a known resistance
3. **D.C. potentiometer**
4. none of the above

Q43. A phase shifting transformer is used in conjunction with ……..

1. D.C. potentiometer
2. **Drysdale potentiometer**
3. A.C. co-ordinate potentiometer
4. Crompton potentiometer

Q44. In order to achieve high accuracy, the slide wire of a potentiometer should be ……

1. **as long as possible**
2. as short as possible
3. neither too small nor too large
4. very thick

Q45. The stator of phase shifting transformer for use in conjunction with an A.C. potentiometer usually has a …….

1. single-phase winding
2. **two-phase winding**
3. three-phase winding
4. none of the above

Q46. In an A.C. co-ordinate potentiometer, the currents in the phase and quadrature potentiometer are adjusted to be ……..

1. **out of phase by 90°**
2. out of phase by 180°
3. out of phase by 60°
4. out of phase by 0°

Q48. For measurements on high voltage capacitors, the suitable bridge is …….

1. Wein bridge

2. Modified De Santy's bridge
3. **Schering bridge**
4. none of the above

Q50. Mutual inductance can be measured by using ……

1. Anderson bridge
2. Maxwell's bridge
3. **Heaviside bridge**
4. any of the above

Q51. The full range of audibility in audio frequency oscillator is ……

1. 0 to 20 Hz
2. **20 Hz to 2 kHz**
3. 20 Hz to 20 kHz
4. 20 Hz to 20 MHz

Q52. A liquid crystal display requires …….

1. **An AC drive**
2. A DC drive
3. Both AC and DC drive
4. None of these

Q53. The detectors used in optical sensors are …….

1. Photodiodes
2. Phototransistors
3. Laser
4. **Only (1) and (2)**
5. All the above

Q54. Optical sensors used for the displacement measurement works on the principal that …….

1. Intensity of light increases with distance
2. **Intensity of light decreases with distance**
3. Intensity of light remains constant with distance
4. Intensity of light increases with time

Q55. Capacitance sensor can measure very small displacement. It can be formed by varying ……

1. Separation
2. Area
3. Permittivity
4. **Either (1) or (2) or (3)**

Q56. The ionization gauge an instrument used for the measurement of …….

1. **Very low pressure**
2. Medium pressure
3. High pressure
4. Very high pressure

Q57. Dead weight gauge is used for the measurement of pressure of ……

1. About 1000 bar
2. About 2000 bar
3. About 5000 bar
4. **About 7000 bar**

Q59. The resistances of potential transformer winding is minimized by using …….

1. **Thick conductors and small length of turns**
2. Thin conductors and small length of turns
3. Thin conductors and large length of turns
4. Thick conductors and large length of turns

Q60. The nominal ratio for a current transformer is given by ……

1. **rated primary winding current/rated secondary winding current**
2. number of turns in the primary winding/number of turns in the secondary winding
3. number of turns in the secondary winding/number of turns in the primary winding
4. rated secondary winding current/rated primary winding current

Q61. Moving iron power factor meter are suitable for 3 phase balanced circuits. It consists of ……..

1. One control spring
2. Two control spring
3. Three control spring
4. **No control spring**

Ans. 4

Q62. If the frequency of electrodynamic power factor meter is doubled then its reading will become……

1. Twice of the original reading
2. Half of the original reading
3. Four times of the original reading
4. **Remains unaffected**

Q63. A moving iron frequency meter consists of …….

1. Two inductive circuits connected in parallel
2. **One inductive and one non inductive circuit connected in parallel**
3. Two non inductive circuits connected in parallel
4. One inductive and one non inductive circuit connected in series

Q64. The electrodynamic frequency meters have …….

1. **Linear scale and their readings does not depends on voltage**
2. Linear scale and their readings depends on voltage
3. Non linear scale and their readings does not depends on voltage
4. Non linear scale and their readings depends on voltage

Q65. The earth resistance can be measured by ……..

1. Fall of potential method
2. Using an earth tester
3. Ducter ohmmeter method
4. **Only (1) and (2)**
5. All the above

Q66. Electrical equipments are generally earthed through an electrode to avoid shocks when someone touches the body of the equipment. The earth resistance is effected by ……..

1. Depth of electrodes buried in the soil
2. Shape and material of earth electrodes
3. Specific resistance of the soil surrounding the electrode
4. **All of these**
5. None of these

Ans. 4

Q67. When a voltmeter – ammeter method is applied for the measurement of resistance, the voltmeter reads a value of 8.28 V and the ammeter reading is 4.14 mA. Then the value of the resistance will be ……

1. 2 kΩ
2. 20 kΩ
3. **200 kΩ**
4. 2000 kΩ

Q68. In AC bridges, the Wagner earth devices are used to …….

1. Remove all the earth capacitances from the bridge circuit
2. Remove harmonics
3. Reduce error caused by stray electric field
4. **All of these**

Q69. Shielding of the capacitor is done to ……

1. Make the value of capacitor definite
2. Balance the bridge without any problem
3. **Both (1) and (2)**
4. None of these

Q70. The indicating instruments with linear scale is …….

1. PMMC
2. Electrostatic instrument
3. Dynamometer instrument
4. Thermocouple instrument

Q71. The resistance potential divider method and capacitance potential divider method is used for…….

1. Both AC and DC
2. **Former method can be used for both AC and DC and the later method can be used only for AC**
3. Former method can be used for AC only and the later method can be used for both AC and DC
4. Former method can be used for DC only and the later method can be used only for AC

Q72. The range of electrostatic voltmeter can be extended by using ……..

1. Resistance potential divider method
2. Capacitance potential divider method
3. **Both (1) and (2)**
4. None of these

Q73. The multiplying factor of electrostatic voltmeters is given by …….

1. **$(C + C_v) / C$**
2. $(C + C_v) / C_v$
3. $C / (C + C_v)$
4. $C_v / (C + C_v)$

Q74. In electrostatic instruments iron is not used for construction. These instruments are ……..

1. Free from hysteresis and eddy current losses
2. Free from temperature errors
3. Dependent on temperature errors
4. **Both (1) and (2)**
5. None of the above

Q75. If an electrostatic voltmeter is used on AC circuit and has non uniform waves, then it will read ……..

1. Average values
2. **RMS values**
3. Peak values
4. All of these

Q76. Electrostatic voltmeter instruments are suitable for ……

1. AC work only
2. DC work only
3. **Both AC and DC work**
4. None of these

Q77. A Kelvin's multicellular voltmeter has a torsion head and a coach spring for …….

1. Protection against accidental fraction of suspension due to vibration
2. For zero adjustment
3. **Torsion head for zero adjustment and coach spring for Protection against accidental fraction of suspension due to vibration**
4. Torsion head for Protection against accidental fraction of suspension due to vibration and coach spring for zero adjustment

Q78. In electrostatic voltmeters, the principle of their operation is the force of attraction between electric charges on neighboring plates between which potential difference is maintained. The attracted – disc type electrostatic instruments are used for the measurement of ……..

1. Very low voltages
2. Low voltages
3. High voltages
4. **Very high voltages**

Q79. Electrostatic instruments are generally used as ……

1. **Voltmeters**
2. Ammeters
3. Wattmeters
4. Watt-hour meters

Q80. If the quantity to be measured remains constant during the process of taking the repeated measurements then the random errors can be eliminated by ……

1. Calculating the mean of the number of repeated measurements
2. Calculating the median of the number of repeated measurements

3. Calculating the sum of the numbers of repeated measurements
4. **Either (1) or (2)**

Q81. The error between the mean of finite data set and mean of infinite data set is known as …….

1. True error of the mean
2. **Standard error of the mean**
3. Finite error
4. Infinite error

Q82. In a measurement system,

1. A single measurement components may have both random errors and systematic errors
2. A measurement system consists of several components with each component having separate errors
3. **Both the statement (1) & (2) are true**
4. Neither statement (1) nor statement (2) is true

Q83. If the degree of damping of an instrument should be adjusted to a value which is sufficient to enable the pointer to rise quickly to its deflected position without overshooting is called as ……

1. Overdamped
2. **Dead beat**
3. Underdamped
4. None of these

Q84. Due to overdamping, the instrument will become …….

1. Slow
2. Lethargic
3. Fast
4. **Both (1) and (2)**

Q85. While measuring resistance by the voltmeter – ammeter method, the maximum possible percentage error in the voltmeter and ammeter are ± 1.8% and ± 1.2% respectively. Then the maximum possible percentage error in the value of resistance will be …….

1. ± 3%
2. ± 4%
3. ± 4.2%
4. ± 4.8%

Q86. If the resistance in a circuit is given by 80 Ω ± 0.2% and the current flowing through it is 5A ± 0.1%, then the uncertainty in the power will be …….

1. ± 0.2 %
2. **± 0.4 %**
3. ± 0.6 %
4. ± 0.8 %

Q87. When a 100 V moving iron voltmeter is of accuracy class 1 – 0 is used in a circuit, it reads 50 V. Then the maximum possible percentage error in the reading is ……

1. 1 %
2. **2 %**
3. 2.5 %
4. 3 %

Q88. In liquid crystal displays, the liquid crystal exhibits properties of ……

1. Liquid
2. Solids
3. Gases
4. **Both (1) and (2)**

Q89. In light emitting diode, the available light emitting region is …….

1. Less than 2.5 mm
2. **From 2.5 to 25 mm**
3. Greater than 25 mm
4. Greater than 50 mm

Q90. Resolver works on the principal of mutual inductance variation. It is mainly used for the measurement of ……

1. Linear displacement
2. Non – linear displacement
3. **Rotary motion**
4. All of these

Q91. In rotary variable differential transformer, the mutual inductance between the primary and secondary coils varies ……

1. **Linearly with the angular displacement**
2. Non – linearly with the angular displacement
3. Linearly with the linear displacement
4. Non – linearly with the linear displacement

Q92. LVDT which is an instrument for the measurement of displacement, works on the principal of ……..

1. Linear inductance

2. Non – linear inductance
3. **Mutual inductance**
4. Linear capacitance

Q93. The instruments used for the measurement of pressure is / are …….

1. Bellows
2. Diaphragms
3. Fiber optic pressure sensors
4. **All of these**
5. None of the above

Q94. A capacitive pressure sensor has a typical measurement uncertainty of …….

1. One
2. Two
3. Three
4. **Four**

Q95. If an instrument transformer is used to extend the ranges of AC instrument, then its reading will depend on ……..

1. R
2. L
3. C
4. All of these
5. **None of these**

Q96. The potential transformers are used to measure large voltage using ……..

1. High range voltmeter
2. Low range voltmeter
3. High range ammeter
4. **Low range ammeter**

Q97. Vibration galvanometer are generally used ……

1. For measuring electric charges
2. **As null – point detectors in ac bridges**
3. As null – point detectors in dc bridges
4. For measuring power

Q98. The two wattmeters used for the measurement of power input read 50 kW each. What will be the readings of the two wattmeters if the power factor is changed to 0.8 leading keeping the total input power same?

1. 28.35W, 71.65W

2. 31.25W, 73.71W
3. 33.33W, 73.33W
4. 38.35W, 75.5W

Ans. 1

Q99. For the measurement of unknown inductance in terms of known capacitance, the suitable ac bridges are ……..

1. Maxwell and Schering bridge
2. Maxwell and Wien's bridge
3. **Maxwell and Hay's bridge**
4. Hay's and Wien's bridge

Q100. The Wien's bridges is suitable for the measurement of frequency of the range of …….

1. Less than 100 Hz
2. **100 Hz to 100 kHz**
3. 1 kHz to 100 MHz
4. More than 100 MHz

Q101. For the measurement of low resistances, Kelvin's double bridge has high accuracy because:

1. **It has two set of ratio arms which eliminates effect of resistance of connecting lead**
2. It has a null indicating galvanometer
3. It has two null indicator
4. It has four sets of ratio arms which eliminates the effect of resistance of connecting lead

Q102. Swamping resistance is a resistance made up of ……

1. **Alloy of manganin and copper**
2. Alloy of aluminium and copper
3. Alloy of nickel and cobalt
4. Alloy of manganin and aluminium

Q103. In a moving coil of a meter swamping resistance is added to …….

1. Reduce the frequency error
2. **Reduce the temperature error**
3. Reduce the power consumption
4. All of these

Q104. The frequency errors in induction instruments can be compensated by the use of ……..

1. Non inductive shunt in both ammeters and voltmeters

2. **Non inductive shunt in case of ammeters and are generally self compensated in case of voltmeters**
3. Self compensated in case of both ammeters and voltmeters
4. Self compensated in case of ammeters and non inductive shunt in case of voltmeters

Q105. For reducing the errors in an induction instrument the alternating current to be measured has ……

1. **Same frequency with which the instrument was calibrated**
2. High frequency compared with which the instrument was calibrated
3. Low frequency compared with which the instrument was calibrated
4. None of these

Q106. In induction voltmeter, split phase windings are obtained by connecting a …….

1. High resistances in series with windings of both the magnets
2. **High resistance in series with the winding of one magnet and an inductive coil in series with the windings of other magnet**
3. An inductive coil in series with the winding of one magnet and a capacitance in series with the windings of other magnet
4. Inductive coils in series with the winding of both the magnets

Q107. A cylindrical type with split phase winding induction ammeters employs ……..

1. A rotating disc
2. **A hollow aluminum drum**
3. A single flux producing winding
4. Either (1) or (2)

Q108. Induction type instruments are generally used as …….

1. Ammeter
2. Voltmeter
3. Wattmeter
4. **All of these**
5. None of these

Q109. Damping torque is the torque which acts on ……..

1. Stationary system of the instrument
2. Moving system of the instrument only when it is stationary
3. **Moving system of the instrument only when it is moving**
4. Stationary system of the instrument only when the moving system is moving

Q110. The gravity controlled instruments has to be kept ……

1. **Vertical**
2. Horizontal
3. Inclined at 45 degree
4. Inclined at 75 degree

Q111. Random errors in a measurement system are due to …….

1. **Environmental changes**
2. Use of uncalibrated instrument
3. Poor cabling practices
4. Unpredictable effects

Q112. Calibration of instrument is an important consideration in measurement system. The errors due to instruments being out of calibration can be rectified by …….

1. **Increasing the frequency of recalibration**
2. Increasing the temperature coefficient
3. Increasing the susceptibility of measuring instrument
4. Decreasing the frequency of recalibration

Q112. The undesirable characteristics of a measuring system is …….

1. Drift
2. Dead zone
3. Non linearity
4. **All of these**

Ans. 4

Q113. If the instrument is used in wrong manner while application, then it will results in …….

1. Systematic error
2. **Instrument error**
3. Random error
4. Environmental error

Q114. Suitable method for the reduction of systematic errors is / are ……

1. Instrument must be designed carefully
2. By introducing an equal and opposite environmental input for compensating the effect of environmental input in a measurement system
3. By adding high gain feedback to measurement system
4. **All of these**

Q115. The systematic errors of an instrument can be reduced by making …….

1. **The sensitivity of instrument to environmental input as low as possible**
2. The sensitivity of instrument to environmental input as high as possible
3. Systematic errors does not depend on the sensitivity of instrument
4. None of these

Q116. The magnitude of environment – induced variation from the specified calibration condition is quantified by ……

1. Sensitivity drift
2. Zero drift
3. Backlash
4. **Both (1) & (2)**

Q117. In a moving coil voltmeter, the input resistance of the meter can be increased by …….

1. Increasing the number of turns in the coil
2. Decreasing the number of turns in the coil
3. Using the same number of coil turns made up of high resistance material
4. **Both (1) & (3)**

Q118. Depending on whether the display is a numeric or alphanumeric, the segmental display is ……

1. 7 segmental
2. 14 segmental
3. 21 segmental
4. **Either (1) or (2)**
5. None of the above

Q119. The measurements which can be simplified by using X – Y recorders is / are ……..

1. Speed – torque characteristics of motors
2. Regulation curves of power supplies
3. Hysteresis curves
4. **All of these**

Q120. Potentiometer is used for the measurement of ……..

1. Linear displacement
2. Angular displacement
3. Non – linear displacement
4. **Only (1) and (2)**
5. All the above

Q121. The displacement measuring instruments is / are ……

1. Potentiometer
2. LVDT
3. RVDT
4. **All of these**

Q122. In fibre – optic level sensors, the amount of light loss depends on ……..

1. **The proportion of cable that is submerged in the liquid**
2. Amount of light which is reflected back
3. The proportion of cable that is not in the liquid
4. Amount of light which is not reflected back

Q123. Instrument which is capable of discriminating temperature differences of even 0.1 degree Celsius is ……..

1. Fibre – optic level sensors
2. Laser method
3. **Thermography**

Q124. Contact devices used for the measurement of level are ……

1. Increases
2. Decreases
3. **Remains same**
4. None of these

Q125. For the measurement of pressure the instruments used can be …….

1. Mechanical
2. Electro – mechanical
3. Electronic
4. **All of these**
5. None of these

Q126. If at one end, the two wires made of different metals are joined together then a voltage will get produced between the two wires due to difference of temp between the two ends of wires. This effect is observed in ……..

1. **Thermocouples**
2. Thermistors
3. RTD
4. Ultrasonics

Q127. Schering bridges are used for the measurement of ……..

1. Unknown capacitance
2. Dielectric loss
3. Power factor

4. **All of these**
5. None of these

Q128. The AC Bridge which is used for the measurement of frequency is ……

1. Schering bridge
2. **Wien bridge**
3. Hay's bridge
4. Anderson bridge

Q129. If the reading of the two wattmeters is equal and opposite while measuring power in a 3 phase induction motor then the power factor of the load will be …….

1. Unity
2. **Zero**
3. 0.5 lagging
4. 0.5 leading

Q130. The transformer ratio of the transformer depends upon the ……

1. Exciting current
2. Secondary current
3. Power factor of secondary circuit
4. **All of these**

Q131. Primary current in a current transformer is determined by ……

1. **The load on the system**
2. The load on its own secondary
3. The load on its own primary
4. All of these

Q132. The bridge suitable for the measurement of capacitance is / are ……

1. Anderson's bridge
2. Hay's bridge
3. Owen's bridge
4. **None of these**